

COMPLEXITY OF AN ALGORITHM

- The complexity of an algorithm refers to the efficiency of the algorithm in terms of the resources it consumes, such as time and space, as a function of the input size.
- The analysis of algorithmic complexity helps in comparing and selecting algorithms for solving a particular problem.
- Generally, we aim for algorithms with **lower time** and **space complexity**, as they tend to be more efficient and scalable.
- It's important to note that there is a distinction between **the worst-case, best-case, and average-case complexities**.
- While **big O notation** typically represents the **worst-case scenario**, other notations like **big Omega (Ω)** and **big Theta (Θ)** can be used to describe the lower bound and tight bounds, respectively.

There are **two** main types of algorithmic complexity:

✚ **Time Complexity.**

✚ **Space Complexity.**

❖ Time Complexity :

- **Time complexity** is defined in terms of how many times it takes to run a given algorithm, based on the length of the input.
- Time complexity is not a measurement of how much time it takes to execute a particular algorithm because such factors as programming language, operating system, and processing power are also considered.
- Time complexity is a type of computational complexity that describes the time required to execute an algorithm.
- The time complexity of an algorithm is the amount of time it takes for each statement to complete.
- As a result, it is highly dependent on the size of the processed data.
- It also aids in defining an algorithm's effectiveness and evaluating its performance.
- Time complexity is commonly expressed using big O notation (e.g., $O(n)$, $O(n^2)$, $O(\log n)$), where n is the size of the input.

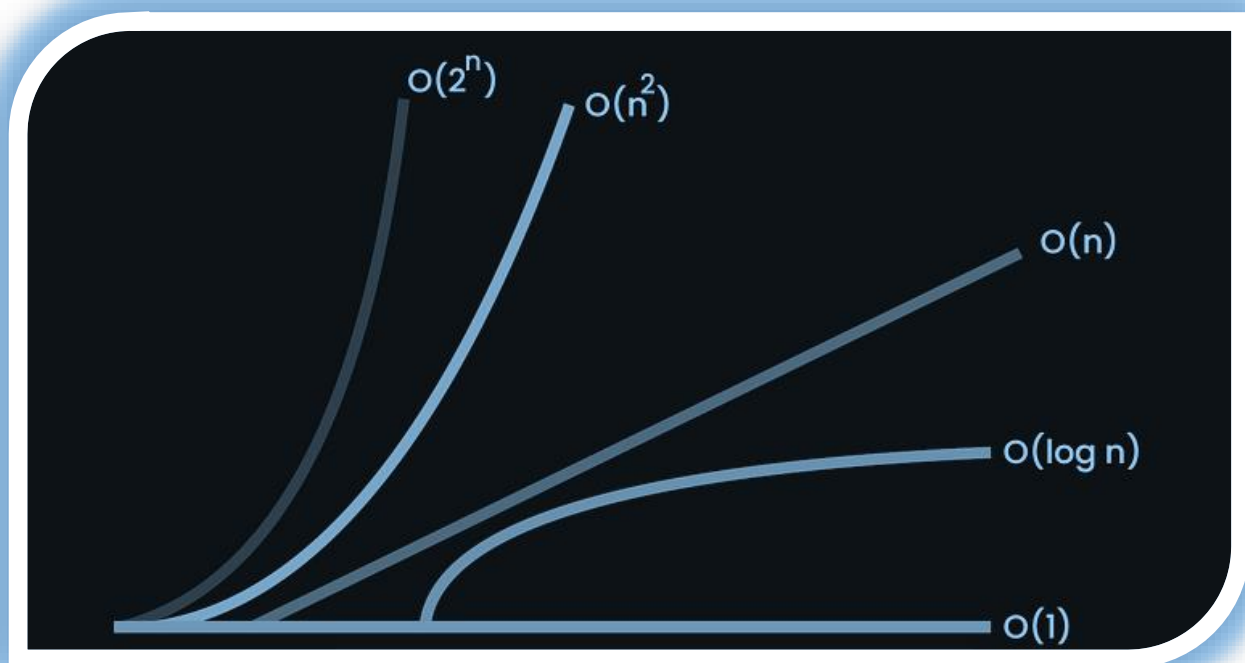
❖ Space Complexity :

- When an algorithm is run on a computer, it necessitates a certain amount of memory space.
- The amount of memory used by a program to execute it is represented by its space complexity.

- Because a program requires memory to store input data and temporal values while running, the space complexity is auxiliary and input space.
- It also has best case, average case, and worst case scenarios, often expressed using big-O notation.

Common time complexities (from best to worst):

- **$O(1)$** - Constant time
- **$O(\log n)$** - Logarithmic time
- **$O(n)$** - Linear time
- **$O(n \log n)$** - Linearithmic time
- **$O(n^2)$** - Quadratic time
- **$O(2^n)$** - Exponential time



Time and Space complexity of an Algorithm — simplified with examples